NICHOL FLOWERS

312.316.8282 | ncmflowers@gmail.com | LinkedIn | Portfolio: nicholflowers.github.io/

SUMMARY

Data Science professional specializing in Machine Learning, Generative AI, Statistics and Data Engineering. Experience developing and evaluating models, designing experiments and applying quantitative analysis to complex problems. Combines technical capabilities with extensive experience developing metrics, measuring performance, identifying drivers of outcomes and translating findings into measurable operational improvements.

EDUCATION

UC Berkeley - Master of Information and Data Science (MIDS) - May 2026
Texas Southern University - Bachelor of Science, Chemistry

TECHNICAL SKILLS

Python, SQL, R, Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Engineering, LLM Evaluation, LangChain, Qdrant, RAGAS, Hugging Face, Machine Learning, Deep Learning, Neural Networks, TensorFlow, Computer Vision, Reinforcement Learning, Model Evaluation, Data Engineering, Neo4j, AWS, Docker, Git, Data Visualization, Statistics, D3.js, Tableau

PROJECTS

RAG for Multiple Audiences, UC Berkeley School of Information

- Built a dual-audience RAG pipeline in LangChain and evaluated embeddings, chunking strategies, retrieval configurations, reranking and generative models.
- Developed an evaluation framework using LLM-as-a-Judge and retrieval-based metrics.

Eye2Voice: Gaze-to-Speech, UC Berkeley School of Information

- Developed a real-time gaze-to-speech system using deep learning and computer vision to enable individuals with severe physical disabilities to communicate through eye movements.

- Won the **Hal R. Varian Award** for top Data Science Capstone project.

Refugee Movement Network Analysis, UC Berkeley School of Information

- Analyzed UN refugee migration data using Neo4j, degree centrality and minimum spanning trees to identify global migration patterns.

Predicting California Wildfires, UC Berkeley School of Information

- Built an 8-class wildfire occurrence classifier using 40+ years of California weather and fire data, predicting across 3-, 7- and 14-day forecast horizons.
- Engineered 100+ geospatial and temporal features using GeoPandas and tuned XGBoost with temporal validation, achieving up to 4× the accuracy of a random baseline.

PROFESSIONAL EXPERIENCE

Cloud Engineer/ AI Development Lead *2021 - Present*
Cisco Systems Remote

- Lead development of an LLM-powered application that processes unstructured incident data, synthesizes information into structured retrospective reports, and generates context-aware questions to identify potential gaps and inconsistencies, reducing preparation time from 5–7 days to under 4 hours.
- Design and refine LLM prompts and guardrails to improve output quality, consistency and reliability while enforcing requirements for content, tone, and clarity.
- Facilitate technical incident retrospectives, challenging engineering teams to identify systemic weaknesses and implement corrective actions that prevent recurrence or mitigate future service impact.

Volunteer Research Associate *2017 - 2018*
University of Illinois - Chicago Chicago, IL

- Improved data quality and consistency for federally funded nutrition and public health research.

Global Problem Manager *2012 - 2017*
HERE Technologies (formerly Navteq & Nokia) Chicago, IL

- Developed operational performance metrics and targets using historical trends and customer requirements to measure service performance and identify improvement opportunities.
- Led the automation of Problem Management reporting, developing real-time dashboards to track performance against defined metrics and targets.
- Improved IT service availability by 35.8% by reducing the volume and impact of service disruptions through root cause analysis.

Process Improvement Lead *2010 - 2012*
Navteq (formerly Nokia, currently HERE) Chicago, IL

- Pioneered Navteq's first product-specific strategy for map data collection and updates, contributing to $8.5 million in cost savings across the product portfolio.
- Optimized map change detection processes reducing manual data collection time by 50%.